

MathConstruct Smoke Test Report

3 Problems · 3 Models · CoT vs Tool Calling (ReAct)

Problems: BxMO 2011 P1 (Number Theory) · Dutch MO 2010 P4 (Mixed) · IMO SL 2018 C1 (Combinatorics)

Models: Qwen3-32B · Llama-3.1-8B · Llama-3.3-70B

API: HuggingFace Inference Providers



Correct (✓)



Incorrect (X)



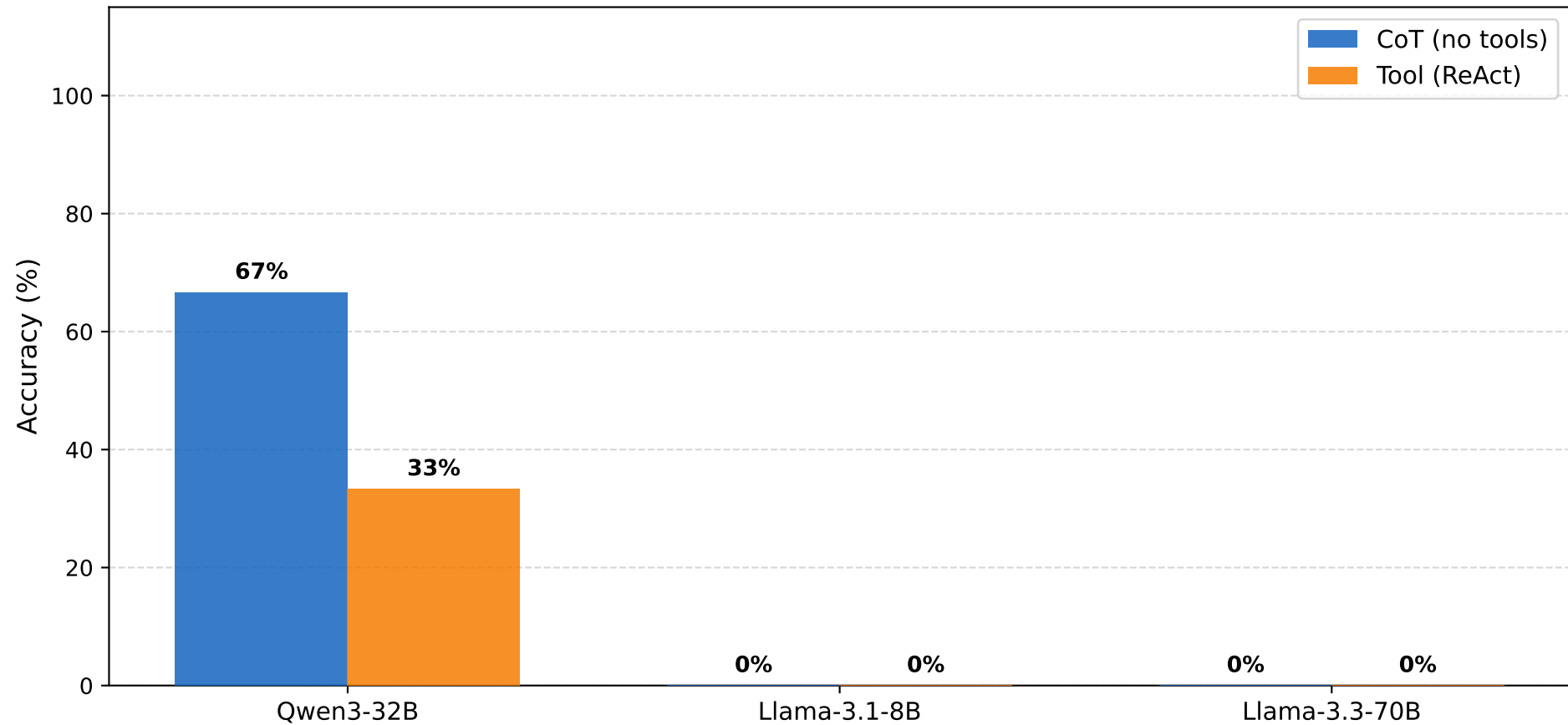
Missing / Error (?)

Correctness by Model × Problem

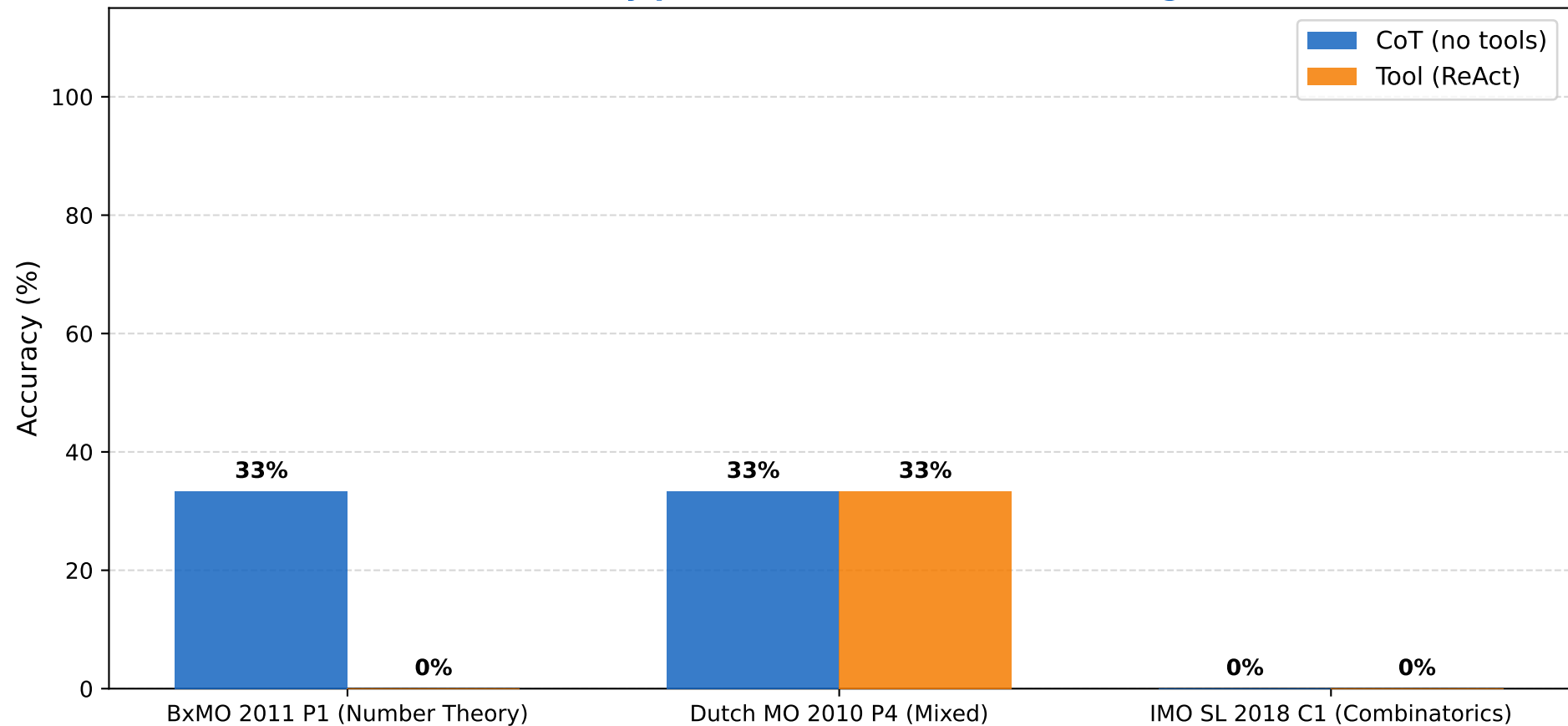
	BxMO 2011 P1 (Number Theory)	Dutch MO 2010 P4 (Mixed)	IMO SL 2018 C1 (Combinatorics)
CoT (no tools)			
Qwen3-32B	✓	✓	✗
Llama-3.1-8B	✗	✗	✗
Llama-3.3-70B	✗	✗	✗

	BxMO 2011 P1 (Number Theory)	Dutch MO 2010 P4 (Mixed)	IMO SL 2018 C1 (Combinatorics)
Tool (ReAct)			
Qwen3-32B	✗	✓	✗
Llama-3.1-8B	✗	✗	✗
Llama-3.3-70B	✗	✗	✗

Accuracy per Model: CoT vs Tool Calling



Accuracy per Problem: CoT vs Tool Calling



Estimated API Cost (USD)

Model	CoT (no tools)	Tool (ReAct)	Total
Qwen3-32B	\$0.0563	\$0.0000	\$0.0563
Llama-3.1-8B	\$0.0042	\$0.0000	\$0.0042
Llama-3.3-70B	\$0.0035	\$0.0000	\$0.0035
TOTAL	\$0.0640	\$0.0000	\$0.0640